BIOE60005 - Bioengineering Group Project

Near-Infrared Silent Speech Interface System Literature Highlights

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SpeeChin: A Smart Necklace for Silent Speech Recognition [1]

This paper presents a wearable silent speech recognition system SpeeChin, which can recognize 54 English and 44 Chinese silent speech commands.

In this system, chin-view images of skin deformation and muscle motion are firstly captured by customized infrared (IR) imaging system mounted on the necklace, including an OV5647 IR camera, 850 nm illumination LED and an IR bandpass optical filter. These under-chin IR image sequences are first pre-processed via skin segmentation, head-pose rectification, motion-enhancing frame differencing, chin-centred cropping and automatic utterance detection, and are then fed to a convolutional–recurrent neural network (CRNN) in which a ResNet-based CNN extracts frame-wise features and an LSTM module models temporal dynamics to classify each sequence into a silent speech command.

On a dataset of 20 participants, SpeeChin recognises 54 English and 44 Chinese silent commands with user-dependent accuracies of about 91 % and achieves over 80 % after user-adaptive fine-tuning in a user-independent setting. The system is compact and socially acceptable but remains limited to command vocabularies and is sensitive to strong sunlight and large head motion.

Key Relevance to our NIR Silent Speech Interface System:

1. SpeeChin shows a full IR image sequence to text label pipeline, confirming that under-chin IR video can support silent speech recognition rather than just generic neck-motion sensing.
2. Its pipeline of skin segmentation, pose rectification, motion-enhancing frame differencing, chin-centred cropping and utterance detection is a ready template for turning our raw NIR neck/throat videos into stable motion features.
3. The CNN + LSTM (CRNN) architecture provides a natural baseline for our NIR-to-text decoder, against which we can later compare more advanced sequence models (e.g. Transformer/CTC) and richer outputs beyond command-level vocabularies.

Estimating Carotid Pulse and Breathing Rate from Near-infrared Video of the Neck [2]

This paper proposes a contact-free way to measure heart rate (HR) and breathing rate (BR) from near-infrared (NIR) video of the neck using a low-cost camera. By modelling NIR reflectance of the neck, the authors show that intensity changes can be split into global rigid motion (head and breathing) and local deformation over the carotid artery, so that pulse and breathing components can be separated.

The pipeline starts with automatic neck ROI detection and downsampling. For HR, pixel intensities are treated as time series in sliding windows: a common average is used to remove global motion, PCA on the residuals isolates carotid-dominated components, and spectral analysis followed by a Hidden Markov Model (HMM) yields a smooth HR trace. BR is estimated by averaging a larger neck/upper-chest ROI, applying a respiratory band-pass filter and running the same spectral + HMM procedure.

On recordings from 12 healthy subjects in both bright and dark conditions, the method achieves mean absolute errors of about 0.36 beats/min for HR and 0.24 breaths/min for BR compared with a reference device and remains reasonably robust to low light and partial collar occlusion. Main limitations are the small, homogeneous cohort, the assumption of near-stationary subjects, and weaker performance when carotid motion is small.

Key Relevance to our NIR Silent Speech Interface System:

1. This research demonstrates that very subtle neck motion can be robustly recovered from low-cost NIR video, supporting our use of neck-mounted NIR imaging to capture laryngeal and articulatory movement.
2. The motion-decomposition approach (common average + PCA separating global and local motion) suggests how we might isolate articulatory patterns from respiratory and vascular signals in our NIR sequences.
3. Spectral analysis with HMM smoothing shows how to turn noisy frame-wise signals into stable trajectories, a strategy we can reuse for modelling the temporal dynamics in NIR-to-text decoding.

Camera Measurement of Physiological Vital Signs [3]

This paper reviews how cameras (RGB, NIR, thermal) can be used to measure physiological signals in a fully non-contact way. It connects basic skin optics with practical camera design and briefly reviews the main signal-processing approaches for camera-based physiological vital-sign monitoring.

Optically, the paper uses the Dichromatic Reflection Model and Lambert–Beer law to explain how light interacts with skin. A key point is that when moving from visible light into near-infrared, penetration depth into the dermis and underlying tissue increases, as does scattering. Visible green light is best for superficial PPG based on haemoglobin absorption, while NIR sacrifices some PPG amplitude but probes a thicker tissue volume and allows invisible active illumination. The hardware discussion compares RGB and NIR cameras mainly in terms of wavelength band, illumination and frame rate, outlining typical non-contact monitoring setups.

The article summarises a standard signal-processing pipeline that uses ROI selection, spatial averaging, band-pass filtering, source separation and spectral estimation. It also reviews deep-learning models that take short video clips as input and predict either waveforms or vital-sign values. The review concludes that camera-based heart rate and breathing rate are relatively mature, while other physiological markers are still harder to estimate reliably.

Key Relevance to our NIR Silent Speech Interface System:

1. The paper supports using NIR cameras with active illumination for neck imaging, since NIR is sensitive to deeper tissue motion and still works well in low light, which fits our laryngeal silent speech setting.
2. It also points to practical signal processing and deep learning methods that we can adapt to extract subtle neck motion from NIR video before decoding it into text.

NeckFace - Continuously Tracking Full Facial Expressions on Neck-mounted Wearables [4]

This paper introduces the NeckFace which is also a neck-mounted infrared system that recovers full facial expressions and head pose from an under-chin view.

The hardware design is almost the same as SpeeChin, including a OV5647 IR camera and 850 nm illumination, embedded in a necklace or neckband and captures lower-face and neck skin motion. IR frames are thresholded to isolate skin and suppress the background, and basic geometric augmentation is used so the model can tolerate small changes in camera position. NeckNet is built on a ResNet-34 backbone and takes each frame as input, predicting 52 ARKit blendshape coefficients and three head rotation angles, with a loss that places more weight on frames where the expression is stronger.

On data from thirteen participants in sitting, walking and remount sessions, NeckFace reconstructs facial motion with errors that are visually close to ground truth and remains usable after the device is taken off and worn again. The current prototype remains user-dependent and was only evaluated on 13 participants in indoor scenarios.

Key Relevance to our NIR Silent Speech Interface System:

1. It validates that under-chin IR imaging can support high-dimensional facial motion regression in a neck-worn form factor.
2. Compared with SpeeChin, which goes directly from IR video to command labels, NeckFace takes a representation-first approach. It shows that we can first learn a continuous motion representation from neck IR images and then map that representation to text in our own system

References

[1] R. Zhang et al., “SpeeChin: A smart necklace for silent speech recognition,” Proc. ACM Interact. Mob. Wearable Ubiquitous Technol., vol. 5, no. 4, pp. 192:1–192:23, 2021.

[2] W. Chen, J. Hernandez, and R. W. Picard, “Estimating carotid pulse and breathing rate from near-infrared video of the neck,” Physiol. Meas., vol. 39, no. 10, art. no. 10NT01, 2018.

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